NR4A3 fusion-junction antisense gapmers for extraskeletal myxoid chondrosarcoma: reagents, test articles and a pre-registrable knockdown experiment

Tristan D. McRae

Independent researcher, unaffiliated. Correspondence: trimcrae@gmail.com ORCID: 0000-0002-1823-1451

ABSTRACT

Extraskeletal myxoid chondrosarcoma (EMC) is an ultra-rare sarcoma usually defined by an in-frame *NR4A3* fusion. That junction is in no normal transcript, so an antisense gapmer could in principle cleave it, sparing its parents; none is reported for any *NR4A3* fusion in the literature retrieved. This computational work performs the in-silico half of step one of an off-target framework; every sequence named is a research reagent not for administration. It names what a laboratory needs: two reagents at the most-reported breakpoints, 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6, both to *NR4A3* exon 3, longest wild-type parent gap duplexes eight and nine base pairs; two screened controls; a pre-registrable selectivity threshold. Both come from a panel of 190 junction-spanning 16-mers tiled 5-6-5 across 38 in-frame junctions: 87 let a mature wild-type parent pair their whole catalytic gap over ten or more contiguous base pairs, and for 61 the longest is wild-type *NR4A3*. Ten is a convention, not a measurement: exon-terminus chimeras meet the same screen at 40.6% against the panel's 45.8%. Five test articles are named; the two fusion-positive EMC cell models are reported at an *NR4A3* exon-2 acceptor, not this panel's. The design procedure is released.

Keywords antisense oligonucleotide; gapmer; RNase-H1; fusion transcript; NR4A3; extraskeletal myxoid chondrosarcoma; off-target screening

Introduction

Most EMC carries an in-frame *EWSR1*::*NR4A3* fusion[1]; *TAF15* is a substantial minority partner and *TCF12* and *TFG* are rare[2]. *FUS*, reported in two of five variant cases in a recent series[3], supplies eight modelled junctions here. Conventional cytotoxic chemotherapy has limited activity[4], though responses occur[5]. Disease control exceeded response with a trialled tyrosine-kinase inhibitor: this interpretation uses the review's response categories[4], not a figure stated in the trial report[6].

A fusion-junction gapmer could recruit RNase-H1 to cleave tumour RNA while sparing the normal parents. Junction-directed nucleic-acid agents have been reported against at least six fusion oncogenes: two as antisense oligonucleotides and the remainder as RNA-interference agents, including one expressed from a lentiviral vector[7-12]. None against an *NR4A3* fusion was found in the literature retrieved.

Each parent supplies roughly half the junction sequence. Such matches often fall outside a conventional off-target search's mismatch budget but may pair the catalytic gap. We therefore screen that pairing directly. RNase-H1's reported minimum DNA gap is six nucleotides, with seven to ten the working range[13]. Our criterion instead concerns a ten-base-pair contiguous duplex through the whole six-nucleotide gap. Ten is an adopted hybridisation criterion, not a measured cleavage threshold or a gap-length requirement. We also adopt, without establishing, the premises that incomplete overall pairing can permit cleavage and that wing mismatches are less protective than gap mismatches. Of the five steps in a 2025 industry off-target framework[14], this work performs only the in-silico half of step one; transcriptomics is not performed. The Discussion proposes a parent-selectivity measurement related to step three.

Materials and Methods

All analyses use public data; no laboratory work was performed. Complete parameters, per-design tables and claim bounds are archived. Canonical transcripts of five partners and *NR4A3* were obtained from Ensembl[15]. We tiled 16-mers in 5-6-5 β-D-oxy-locked-nucleic-acid/DNA/β-D-oxy-locked-nucleic-acid geometry[13], giving five registers per junction with the breakpoint inside the DNA gap. Six is the cited minimum, not the preferred gap length; the genome-wide arm is available only for 16-mers. Gap-level margin counts junction-unique gap bases on the shorter side of the breakpoint.

Five screens examine human RefSeq RNA alignments; an exhaustive transcript substitution search within one mismatch; unspliced parents; the longest contiguous duplex through the gap in six mature wild-type parents; and unambiguous GRCh38, including mitochondrial sequence. This adopted scope covers mature, precursor, exon-exon, non-coding and mitochondrial sequence. A near-match pairs at least 14 of 16 positions. Parent liability requires a contiguous run of at least ten base pairs pairing the whole gap. Ten identically screened null ensembles comprise four shuffles, two base-frequency draws (uniform or composition-matched), and four real-parent chimeras, two joined at exon termini.

Alignments were re-scored by nearest-neighbour stability of the longest contiguous paired run; only energy separations are reported. Melting-temperature calculations assume unmodified DNA:RNA at 250 nM strand[16], not locked phosphorothioate chemistry. Accordingly, we report no absolute melting temperature for the proposed reagents. Table 1 gives the unmodified-hybrid model's fusion-versus-parent difference. LNA and phosphorothioate effects are unmodelled; this difference is not a validated bound for the modified chemistry.

Every value in the tables below is read from a committed artifact and none is typed by hand: the reagents from `fusion-junction-aso-sequences.csv`, the canonical machine-readable record, and the controls from `aso-control-oligos.json`, either directly or composed from their fields. The captions carry literature facts, each with its source named there. Every reagent named here is a 5-6-5 phosphorothioate gapmer. An oligonucleotide should be ordered from that file rather than transcribed from this page.

Table 1. The two reagents named for synthesis, with their parent-duplex label. The parent-duplex column is the longest contiguous duplex a mature wild-type parent forms through the catalytic gap; neither reagent reaches the ten-base-pair criterion, so the length is printed rather than a pass mark. The test articles are the engineered constructs of Brenca et al. (PMID: 31020999) — E-N for the *EWSR1* reagent and T-N* for the *TAF15* reagent; the two patient-derived models of Bangerter et al. (PMID: 36316541) are reported at an NR4A3 exon-2 acceptor; correspondence to these reagents requires nucleotide-junction confirmation. ΔTm separates the fusion duplex from the more stable half of the design's own target window, which is a different parent from the duplex column's searched wild-type TFG. These are differences from the unmodified DNA:RNA model at 250 nM strand concentration. LNA and phosphorothioate effects are unmodelled, so these values are not validated predictions or bounds for the proposed modified reagents. Nothing here has been synthesised or tested, and no sequence may be administered to any person or animal.

Table 1

seam	reagent	margin	WT gap duplex (bp)	Model ΔTm (°C)
EWSR1 e12::*NR4A3* e3	5′-GGGCATATCATCAAAC-3′	3	8 bp, wild-type *TFG*	26.6
TAF15 e6::*NR4A3* e3	5′-GGGCATATCTTGTGTG-3′	3	9 bp, wild-type *TFG*	36.0

Results

The reagents

We select the highest-margin designs at the two most-prevalent junctions with published exon-resolved breakpoints: 5′-GGGCATATCATCAAAC-3′ for *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ for *TAF15* exon 6, both to *NR4A3* exon 3 (Table 1). Exon indices count from the transcript 5′ end, including non-coding exons; coding-exon numbering can select a different reagent. Both have gap-level margin three.

Their longest wild-type duplexes pairing the whole gap are eight and nine base pairs, respectively, both against *TFG*. Thus both are liable at an eight-base-pair cut, the *TAF15* reagent remains liable at nine, and neither is liable at ten. Both also pair part of the gap at the wild-type *NR4A3* exon-2/exon-3 seam; these unmeasured partial duplexes are not counted. A one-base register shift can reverse the verdict: 5′-AGGGCATATCTTGTGT-3′, adjacent to the *TAF15* reagent, pairs 11 base pairs of wild-type *NR4A3* through the whole gap and cannot substitute for it.

At a deeper-than-default search ceiling, the *EWSR1* and *TAF15* reagents have 123 and eight gap-paired sense-strand transcript near-matches, but six and five gene loci. Most of the 123 are predicted transcript models. The *EWSR1* reagent also matches wild-type *TAF15* precursor RNA across an intron-exon boundary with two mismatches, one in the gap. Ten shared donor bases let this reagent span *EWSR1*, *TAF15* and *FUS* breakpoints (Figure 1). The *TAF15* reagent has no sense-strand precursor site.

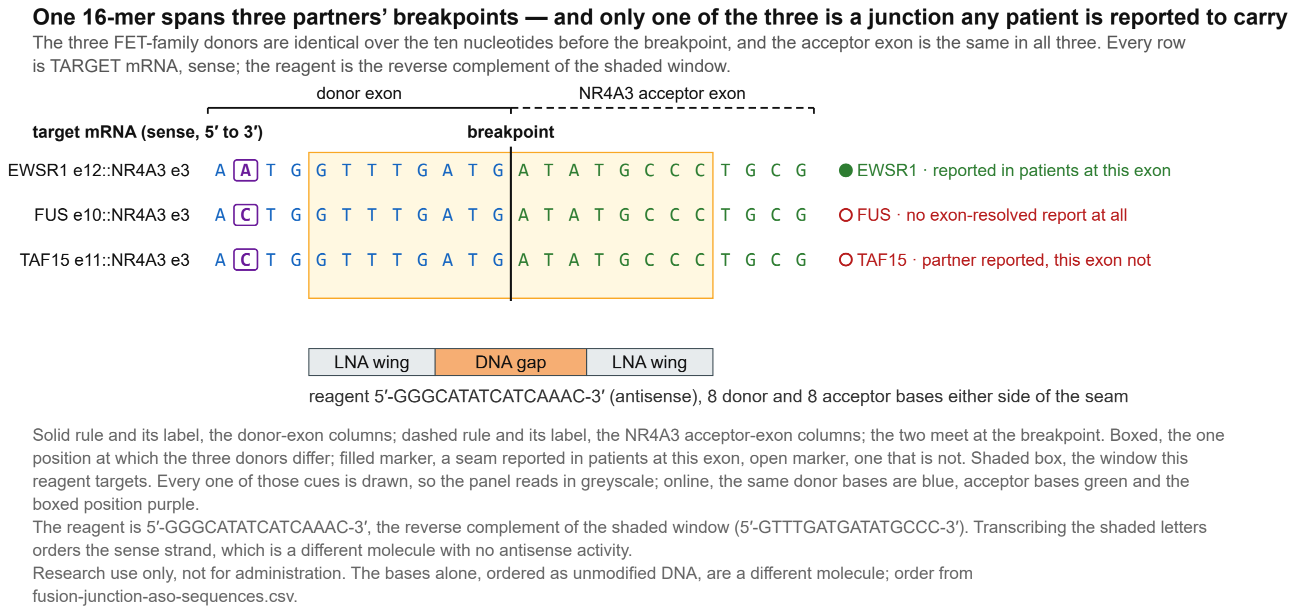

Figure 1. One 16-mer spans three partners' breakpoints; only one is reported in patients. Breakpoint-aligned windows join *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 to *NR4A3* exon 3, with reporting status for each row. The *TAF15* exon-11 row differs from Table 1's exon-6 reagent: it is an additional breakpoint spanned by the *EWSR1* reagent, not a selected reagent.

Aggregate genome-wide loads are similar: 156 and 135 hybridisable gap-paired sites. The *EWSR1* reagent has one exact 16-base genomic match, intronic in annotated lncRNA ENSG00000304430; the *TAF15* reagent has none. These are sequence predictions, not measured activity. Much genomic sequence is untranscribed; transcription and cleavage at these sites are unmeasured.

Both reagents are phosphorothioate throughout, with five contiguous β-D-oxy-locked residues per wing, exceeding the two to four taken here as usual. Their high affinity may permit more mismatches than the near-match screens' two-mismatch ceiling. Both begin with a locked 5′-GGG tract. Increased sequence-dependent hepatotoxicity with high affinity is an adopted, untested premise here, not a retrieved finding. Neither reagent's potency or safety is established.

Combining the breakpoint distribution of an 18-case series[17] with a 58-case cohort[18] assigns 68.4% of molecularly confirmed cases to these two junctions. This is modelled coverage, not patient screening. Its 39.9%–82.8% range is not a confidence interval: two of four inputs are fixed, no nominal level is assigned, and the calculation transfers a breakpoint distribution between cohorts collected 21 years apart. The *TAF15* arm uses three of three reported breakpoints, an upper bound. Adding the top-margin *EWSR1* exon-13/*NR4A3* exon-3 design, 5′-GGGCATATCTCCACGG-3′, would raise modelled coverage to 79.0%; selection names only the first two junctions by reported prevalence.

Selection from a panel of 190 designs

Across 38 in-frame junctions of five partners, 87 of 190 designs let a mature wild-type parent pair the whole gap over at least ten contiguous base pairs. The longest duplex is with *NR4A3* for 61; 85 pair one of their own two parents. Only seven of the 87 reach the conventional 14-of-16 near-match threshold, and parent records are excluded by name from that screen. Nineteen designs have sense-strand precursor near-matches pairing the whole gap. The mature-parent and precursor union is 93 designs, not their sum. Gap mismatches score zero, so these counts bound the fully paired class, not all parent liability.

Alignment-screen cleanliness has additional bounds: seven searches never returned and censoring leaves only 47 of 183 returned searches assessable. A clean count is a floor over that subset; most default-clean designs fail a deeper search. Parent, precursor and genome screens cover all 190 without failures or censoring.

Energy re-scoring finds eight designs with fully paired 16-base off-target duplexes and 45 with a duplex within 2 kcal/mol of the intended target. Neither named reagent is in these classes; their nearest separations are 3.2 and 3.0 kcal/mol. These are upper bounds because the longest-run calculation ignores pairing on either side of a mismatch.

Longer gaps retain parent liability: counts are 87, 88 and 87 in panels of 190, 266 and 342, respectively, reducing rates from 45.8% to 33.1% and 25.4% without removing the problem. At 5-10-5, whole-gap pairing itself requires ten base pairs; shorter runs potentially compatible with the reported seven-to-ten working range are missed, making 25.4% a floor.

Three designs clear every screen at the ten-base-pair cut. Two have no full-gap parent duplex at any length; the third pairs wild-type *NR4A3* over eight bases. None targets a reported patient junction, so these are mechanism controls. Within-junction selection instead finds a parent-clear design for 35 of 38 junctions and all five with published exon-resolved breakpoints. At cuts of nine, eight, seven and six base pairs, those counts become 31/3, 23/2, 9/0 and 6/0 (all junctions/published junctions). These cuts measure whole duplexes, not DNA gap length; the named *TAF15* junction loses its best design at nine.

Exon-terminus chimeras meet the parent-liability criterion at 40.6%, versus 45.8% for the panel. The strongest null falls inside the panel's 95% interval at cuts seven through thirteen except eleven; the excess changes sign four times across cuts six through thirteen. Both panel and chimeras contain mostly junctions unreported in patients. The comparison therefore does not resolve a disease-specific excess over the liability generated by joining these genes' exon termini.

Test articles

Five test articles offer complementary routes. The panel joins donors to *NR4A3* exon 3, its first coding exon, whereas two patient-derived cell models are reported at exon 2. No annotated *NR4A3* transcript has exon 2 as its first coding exon, so annotation does not reconcile them. The original protein-fusion filter excluded acceptors upstream of the initiation codon; that filter is inappropriate for RNase-H1 gapmers, which cleave RNA independently of reading frame. An *EWSR1* exon-7/*NR4A3* exon-2 fusion was sequenced in one of five *EWSR1*-rearranged tumours in a transcriptome series[19], and a *PGR*::*NR4A3* case joins exon 2 to the *NR4A3* 5′ untranslated region[20]. All retrieved exon-resolved acceptors are exon 2, exon 3 or a cryptic exon in intron 2[21]; none is downstream of exon 3, where no patient-grounded designs are made.

Three engineered constructs come from a functional study reporting their exon spans[21]. Two, E-N and T-N*, carry exactly the named reagents' junctions. Rebuilding them offers a direct test of junction-selective knockdown, but heterologous complementary-DNA overexpression cannot establish activity at an endogenous locus.

The two patient-derived, identity-clean models are USZ20-EMC1 (RRID:CVCL_C6MX) and USZ22-EMC2 (RRID:CVCL_C6MY), the only fusion-positive EMC cell source identified here[22]. They are available on request, with no repository deposit and reported doubling times of five to six days. Their reported junctions join *EWSR1* exon 13 and *TAF15* exon 6 to *NR4A3* exon 2. The report supplies no sequenced boundary, transcript accession or junction sequence; whether this denotes a non-coding acceptor or unsupported numbering cannot be determined. An earlier version of this work was withdrawn after this class of indexing error.

A first-coding-exon acceptor is more parsimonious for the defining chimeric transcription factor[1,21]. A non-coding exon-2 acceptor would instead retain the *NR4A3* initiation codon under the donor promoter. On the former interpretation, USZ22-EMC2 matches the named *TAF15* reagent and USZ20-EMC1 matches the third, exon-13 design. This remains inference. The builder emits exon-2 acceptors only for published patient seams or user-supplied sequencing checked against its transcript models.

Alternative exon-2 reagents are 5′-AGTGGGCTCTCCACGG-3′ (*EWSR1* exon 13) and 5′-AGTGGGCTCTTGTGTG-3′ (*TAF15* exon 6), both at top margin and below the ten-base-pair criterion. Their longest full-gap parent duplexes are eight bases with *EWSR1* and nine with *NR4A3*, respectively. The latter directly involves the acceptor used in the selectivity ratio. Reagents cannot be exchanged between acceptors.

Establish the test article's breakpoint at nucleotide resolution by RNA sequencing before ordering: only nine of 176 distinct panel sequences match multiple junctions. Routine break-apart *NR4A3* fluorescence in situ hybridisation detects rearrangement regardless of partner[4], not the nucleotide seam.

Controls for the knockdown experiment

Table 2 gives a dinucleotide-preserving scramble of each reagent, screened against mature parents. Among such scrambles, 10.0% pair a parent's whole gap over at least ten bases and 3.9% have the longest duplex with *NR4A3*. Passing this screen is necessary for these controls; it does not establish inertness.

Table 2. The two control oligonucleotides, each screened as its reagent was. Each is a dinucleotide-preserving scramble of the reagent it controls, matching it in length, first and last base, base composition and dinucleotide counts while spanning no junction, and each cleared the same mature-parent screen the reagent did. The Controls section above explains why that screening step makes a scramble a control.

Table 2

control	sequence	scramble of	WT gap duplex (bp)
control-1	5′-GGGCATCAACATAATC-3′	the EWSR1 e12 :: NR4A3 e3 reagent	6 bp, wild-type *TAF15*
control-2	5′-GTATGTCATTGGCTGG-3′	the TAF15 e6 :: NR4A3 e3 reagent	7 bp, wild-type *EWSR1*

Discussion

The falsification experiment

We propose isogenic fusion-positive/fusion-negative comparisons, as used for *NAB2::STAT6* in solitary fibrous tumour[23], with the screened controls. Knockdown alone cannot distinguish the relevant failure modes.

Selectivity is wild-type *NR4A3* knockdown IC50 divided by fusion knockdown IC50, from matched dose responses in the same wells. This ratio and its threshold of 5.0 are adopted conventions, not specified by framework step three[14]. With an assumed replicate standard deviation of 0.35 for the natural-log selectivity ratio, six independent biological replicates give about 80% power to falsify true selectivity of 3; three give about 30%. Variance is unmeasured.

Above a realised standard deviation of approximately 0.65 at three replicates, 1.53 at six or 2.25 at ten, no observed ratio at least one can place a two-sided 95% interval's upper limit below 5. Such a test can fail only with an anti-selective result and is treated as void. A normal-approximation interval would change these figures. Apply the gate to the upper confidence bound on a pilot's standard deviation: at or above the threshold for the proposed replicate count, increase replication or do not test. An acceptor-only ratio cannot exclude other parent liabilities; measure *TFG* alongside *NR4A3*, given its eight- and nine-base full-gap duplexes with the named reagents.

Interpretation and limits

Every modelled in-frame junction is designable; discrimination remains unproven. A design's own parents pair at most 13 bases across mature and precursor compartments, whereas eight off-target duplexes pair all 16, five in curated records. This is no liability ranking: the parent screen reads six transcripts and the energy screen excludes parents. Mature-parent liability requires a ten-base full-gap duplex; precursor liability requires a full-gap near-match within two mismatches. These are distinct, overlapping classes.

The archive additionally screens the patient's unrearranged *NR4A3* allele. Its two-mismatch ceiling gives a lower bound. No selected design fails, but two other registers at the *EWSR1* exon-13/*NR4A3* exon-2 seam do, reinforcing the register hazard.

Four cited junction-specificity reports tested synthesised molecules[9-11,24]; we did not survey design pipelines and claim no priority for pre-synthesis screening. Whether sparing wild-type *NR4A3* justifies a specificity cost remains unsettled: reported paralogue redundancy and dosage effects point in opposite directions.

All five screens predict hybridisation, not duplex formation or cleavage. Junction-directed oligonucleotides are established; the new application here is the indication. No potency, therapeutic window or safety has been measured, and systemic delivery to solid tumours remains unresolved. All named test articles require cell culture; none establishes clinical readiness.

Acknowledgments

No person other than the author contributed to this work.

Author Contributions

T.D.M. conceived and directed the project and is responsible for its content. AI assistance with analysis and manuscript preparation is disclosed below.

Statements and Declarations

Research use only, and not for administration to any person or animal. Every sequence is an unsynthesised, untested laboratory reagent. Order only from `fusion-junction-aso-sequences.csv`, which specifies sequence, locked residues and backbone, after RNA sequencing establishes the test article's breakpoint at nucleotide resolution.

Ethical considerations. Not applicable; no human subjects, human material or animals were involved, and no ethics approval was required.

Consent to participate. Not applicable; no participants were enrolled.

Consent for publication. Not applicable; no individual-level data are reported.

Declaration of conflicting interest. No competing financial interests exist. The author holds no patent, patent application, equity or consultancy relating to any sequence or method described here. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Funding statement. No external funding; self-funded by the author.

Use of artificial intelligence. Claude (Anthropic) assisted with analysis code, screens, literature retrieval and checking, and manuscript drafting and review. GPT-6-Astra (OpenAI) assisted with this revision and checks against repository artefacts. Bibliographic records were retrieved from PubMed, Europe PMC or Crossref rather than generated, and citations were checked against those records. The author directed the work and is responsible for its content.

Data availability. Code, per-design tables and screen parameters for the preceding analysis are archived at doi:10.5281/zenodo.22229096. This revision does not claim that the archived files contain its corrected interpretations. The accompanying sequence record (Supplementary File 1) and revision note (Supplementary File 2) correct the melting-temperature interpretation and qualify cell-model correspondence; sequence rows and numerical model outputs are unchanged. An earlier analysis mislocated the acceptor through coding-versus-transcript exon indexing and was withdrawn in full. Rebuilt, verified panels and the correction record are archived.

References

1. Labelle Y, J Zucman, G Stenman, LG Kindblom, J Knight, C Turc-Carel, B Dockhorn-Dworniczak, N Mandahl, C Desmaze, et al. (1995). Oncogenic conversion of a novel orphan nuclear receptor by chromosome translocation. Hum Mol Genet 4:2219–2226. doi:10.1093/hmg/4.12.2219
2. Paioli A, S Stacchiotti, D Campanacci, E Palmerini, AM Frezza, A Longhi, S Radaelli, DM Donati, G Beltrami, et al. (2021). Extraskeletal Myxoid Chondrosarcoma with Molecularly Confirmed Diagnosis: A Multicenter Retrospective Study Within the Italian Sarcoma Group. Ann Surg Oncol 28:1142–1150. doi:10.1245/s10434-020-08737-7
3. Chen X, X He, R Peng, M Chen and H Zhang. (2026). A series of extraskeletal myxoid chondrosarcomas with rare morphological and molecular variations. Histopathology 89:181–188. doi:10.1111/his.70131

4. Remiszewski P, S Falkowski, A Szumera-Ciećkiewicz, MJ Spałek, P Rutkowski and AM Czarnecka. (2025). From pathogenesis to the patient's bedside: a comprehensive review of extraskeletal myxoid chondrosarcoma. J Cancer Res Clin Oncol 151:283. doi:10.1007/s00432-025-06316-5
5. Stacchiotti S, GP Dagrada, R Sanfilippo, T Negri, I Vittimberga, S Ferrari, F Grosso, G Apice, M Tricomi, et al. (2013). Anthracycline-based chemotherapy in extraskeletal myxoid chondrosarcoma: a retrospective study. Clin Sarcoma Res 3:16. doi:10.1186/2045-3329-3-16
6. Stacchiotti S, S Ferrari, A Redondo, N Hindi, E Palmerini, MA Vaz Salgado, AM Frezza, PG Casali, A Gutierrez, et al. (2019). Pazopanib for treatment of advanced extraskeletal myxoid chondrosarcoma: a multicentre, single-arm, phase 2 trial. Lancet Oncol 20:1252–1262. doi:10.1016/S1470-2045(19)30319-5
7. Skórski T, C Szczylik, L Malaguarnera and B Calabretta. (1991). Gene-targeted specific inhibition of chronic myeloid leukemia cell growth by BCR-ABL antisense oligodeoxynucleotides. Folia Histochem Cytobiol 29:85–89.
8. Toretsky JA, Y Connell, L Neckers and NK Bhat. (1997). Inhibition of EWS-FLI-1 fusion protein with antisense oligodeoxynucleotides. J Neurooncol 31:9–16. doi:10.1023/a:1005716926800
9. Parker Kerrigan BC, D Ledbetter, M Kronowitz, L Phillips, J Gumin, A Hossain, J Yang, M Mendt, S Singh, et al. (2020). RNAi technology targeting the FGFR3-TACC3 fusion breakpoint: an opportunity for precision medicine. Neurooncol Adv 2:vdaa132. doi:10.1093/noajnl/vdaa132
10. Ward SV, T Sternsdorf and NB Woods. (2011). Targeting expression of the leukemogenic PML-RARα fusion protein by lentiviral vector-mediated small interfering RNA results in leukemic cell differentiation and apoptosis. Hum Gene Ther 22:1593–1598. doi:10.1089/hum.2011.079
11. Shao L, I Tekedereli, J Wang, E Yuca, S Tsang, A Sood, G Lopez-Berestein, B Ozpolat and M Ittmann. (2012). Highly specific targeting of the TMPRSS2/ERG fusion gene using liposomal nanovectors. Clin Cancer Res 18:6648–6657. doi:10.1158/1078-0432.ccr-12-2715
12. Neumayer C, D Ng, D Requena, CS Jiang, A Qureshi, R Vaughan, TP Prakash, A Revenko and SM Simon. (2024). GalNAc-conjugated siRNA targeting the DNAJB1-PRKACA fusion junction in fibrolamellar hepatocellular carcinoma. Mol Ther 32:140–151. doi:10.1016/j.ymthe.2023.11.012
13. Kauppinen S, B Vester and J Wengel. (2005). Locked nucleic acid (LNA): High affinity targeting of RNA for diagnostics and therapeutics. Drug Discov Today Technol 2:287–290. doi:10.1016/j.ddtec.2005.08.012
14. Andersson P, SA Burel, H Estrella, J Foy, PH Hagedorn, TA Harper Jr, SP Henry, JC Hoflack, EM Holgersen, et al. (2025). Assessing Hybridization-Dependent Off-Target Risk for Therapeutic Oligonucleotides: Updated Industry Recommendations. Nucleic Acid Ther 35:16–33. doi:10.1089/nat.2024.0072
15. Dyer SC, O Austine-Orimoloye, AG Azov, M Barba, I Barnes, VP Barrera-Enriquez, A Becker, R Bennett, M Beracochea, et al. (2025). Ensembl 2025. Nucleic Acids Res 53:D948–D957. doi:10.1093/nar/gkae1071
16. Sugimoto N, S Nakano, M Katoh, A Matsumura, H Nakamuta, T Ohmichi, M Yoneyama and M Sasaki. (1995). Thermodynamic parameters to predict stability of RNA/DNA hybrid duplexes. Biochemistry 34:11211–11216. doi:10.1021/bi00035a029
17. Panagopoulos I, F Mertens, M Isaksson, HA Domanski, O Brosjö, S Heim, B Bjerkehagen, R Sciot, P Dal Cin, et al. (2002). Molecular genetic characterization of the EWS/CHN and RBP56/CHN fusion genes in extraskeletal myxoid chondrosarcoma. Genes Chromosomes Cancer 35:340–352. doi:10.1002/gcc.10127
18. Huang SC, JC Lee, YC Hsu, JW Tsai, YC Kao, TH Hsieh, YM Chang, KC Chang, PS Wu, et al. (2023). Extraskeletal Myxoid Chondrosarcomas: The Uncommon Clinicopathologic Manifestations and Significance of TAF15::NR4A3 Fusion. Mod Pathol 36:100161. doi:10.1016/j.modpat.2023.100161
19. Urbini M, V Indio, A Astolfi, G Tarantino, SL Renne, S Pilotti, AP Dei Tos, R Maestro, P Collini, et al. (2018). Identification of an Actionable Mutation of KIT in a Case of Extraskeletal Myxoid Chondrosarcoma. Int J Mol Sci 19:E1855. doi:10.3390/ijms19071855
20. Wilbur HC, DR Robinson, YM Wu, C Kumar-Sinha, AM Chinnaiyan and R Chugh. (2022). Identification of Novel PGR-NR4A3 Fusion in Extraskeletal Myxoid Chondrosarcoma and Resultant Patient Benefit From Tamoxifen Therapy. JCO Precis Oncol 6:e2200039. doi:10.1200/po.22.00039
21. Brenca M, S Stacchiotti, K Fassetta, M Sbaraglia, M Janjusevic, D Racanelli, M Polano, S Rossi, S Brich, et al. (2019). NR4A3 fusion proteins trigger an axon guidance switch that marks the difference between EWSR1 and TAF15 translocated extraskeletal myxoid chondrosarcomas. J Pathol 249:90–101. doi:10.1002/path.5284
22. Bangerter JL, KJ Harnisch, Y Chen, C Hagedorn, L Planas-Paz and C Pauli. (2023). Establishment, characterization and functional testing of two novel ex vivo extraskeletal myxoid chondrosarcoma (EMC) cell models. Hum Cell 36:446–455. doi:10.1007/s13577-022-00818-x
23. Li Y, JT Nguyen, M Ammanamanchi, Z Zhou, EF Harbut, JL Mondaza-Hernandez, CA Meyer, DS Moura, J Martin-Broto, HN Hayenga and L Bleris. (2023). Reduction of Tumor Growth with RNA-Targeting Treatment of the NAB2-STAT6 Fusion Transcript in Solitary Fibrous Tumor Models. Cancers (Basel) 15:3127. doi:10.3390/cancers15123127
24. Lee MS, S An, JY Song, M Sung, K Jung, ES Chang, J Choi, DY Oh, YK Jeon, et al. (2023). Cancer-Specific Sequences in the Diagnosis and Treatment of NUT Carcinoma. Cancer Res Treat 55:452–467. doi:10.4143/crt.2022.910